

## ANGERS AP, France

WMO: 073901

Lat: 47.5619N

Lon: 0.3169W

Elev: 59

StdP: 100.62

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |             |            |             | Coldest Month WS/MCDB |             |            |             | MCWS/PCWD<br>to 99.6% DB |           | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|-------------|------------|-------------|-----------------------|-------------|------------|-------------|--------------------------|-----------|--------------|
|               |             |             | 99.6%                         |            |             | 99%         |            |             | 0.4%                  |             | 1%         |             |                          |           |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP          | HR         | MCDB        | WS                    | MCDB        | WS         | MCDB        | MCWS                     | PCWD      |              |
| (a)<br>1      | (b)<br>-5.0 | (c)<br>-3.2 | (d)<br>-8.1                   | (e)<br>1.9 | (f)<br>-3.4 | (g)<br>-6.0 | (h)<br>2.3 | (i)<br>-2.0 | (j)<br>10.6           | (k)<br>10.8 | (l)<br>9.4 | (m)<br>10.0 | (n)<br>1.5               | (o)<br>60 | (p)<br>0.500 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |           |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|-----------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |           |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD      |
| (a)<br>7      | (b)<br>12.4            | (c)<br>31.2     | (d)<br>20.3 | (e)<br>29.1 | (f)<br>19.3 | (g)<br>27.1 | (h)<br>18.7 | (i)<br>21.4         | (j)<br>28.9 | (k)<br>20.4 | (l)<br>27.0 | (m)<br>19.5 | (n)<br>25.4 | (o)<br>3.3           | (p)<br>60 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 19.1                            | 13.9 | 23.4 | 18.1 | 13.1 | 22.7 | 17.2 | 12.3 | 21.7 | 62.5          | 29.0 | 58.8 | 27.2 | 55.6 | 25.3 | 27.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 9.0               | 7.8  | 6.8 | -7.6                       | 35.3 | 2.3                | 2.4 | -9.2                                               | 37.0 | -10.6      | 38.4 | -11.9      | 39.8 | -13.5      | 41.5 |  |  |
|                   |      |     | -8.0                       | 23.5 | 2.4                | 1.2 | -9.7                                               | 24.3 | -11.1      | 25.0 | -12.4      | 25.6 | -14.2      | 26.5 |  |  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 11.8   | 4.8   | 5.5   | 8.0   | 10.9  | 13.7  | 17.5  | 19.6  | 18.6  | 16.2  | 12.8  | 8.3   | 5.5   |
|                                                                        | DBStd       | 6.22   | 3.90  | 3.82  | 3.16  | 3.34  | 3.22  | 3.66  | 3.30  | 3.08  | 3.02  | 3.59  | 3.60  | 3.75  |
|                                                                        | HDD10.0     | 641    | 166   | 129   | 79    | 27    | 5     | 1     | 0     | 0     | 1     | 15    | 73    | 145   |
|                                                                        | HDD18.3     | 2537   | 421   | 359   | 321   | 224   | 147   | 58    | 24    | 33    | 76    | 174   | 301   | 399   |
|                                                                        | CDD10.0     | 1301   | 3     | 3     | 17    | 54    | 119   | 224   | 297   | 266   | 188   | 102   | 23    | 5     |
|                                                                        | CDD18.3     | 154    | 0     | 0     | 0     | 0     | 4     | 31    | 62    | 41    | 13    | 2     | 0     | 0     |
|                                                                        | CDH23.3     | 1723   | 0     | 0     | 0     | 19    | 57    | 361   | 672   | 451   | 150   | 13    | 0     | 0     |
|                                                                        | CDH26.7     | 558    | 0     | 0     | 0     | 1     | 7     | 123   | 241   | 153   | 31    | 2     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 3.0    | 3.2   | 3.4   | 3.5   | 3.0   | 3.0   | 2.8   | 2.8   | 2.6   | 2.5   | 2.6   | 3.0   | 3.2   |
| (15) Precipitation                                                     | PrecAvg     | 679    | 68    | 53    | 47    | 55    | 60    | 43    | 45    | 45    | 54    | 72    | 70    | 68    |
|                                                                        | PrecMax     | 961    | 139   | 105   | 135   | 158   | 96    | 103   | 126   | 108   | 130   | 195   | 134   | 147   |
|                                                                        | PrecMin     | 476    | 6     | 2     | 5     | 9     | 26    | 6     | 7     | 11    | 1     | 18    | 17    | 12    |
|                                                                        | PrecStd     | 120    | 32    | 29    | 31    | 35    | 21    | 25    | 31    | 30    | 37    | 36    | 31    | 32    |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 13.9  | 16.4  | 20.9  | 25.8  | 27.2  | 34.9  | 35.1  | 34.0  | 29.9  | 24.8  | 18.1  | 14.5  |
|                                                                        |             | MCWB   | 12.3  | 10.9  | 12.6  | 15.8  | 18.8  | 21.7  | 21.4  | 20.8  | 18.5  | 17.8  | 14.6  | 12.4  |
|                                                                        | 2%          | DB     | 13.0  | 13.8  | 18.0  | 23.1  | 25.0  | 29.9  | 31.8  | 30.8  | 27.1  | 22.0  | 16.2  | 13.8  |
|                                                                        |             | MCWB   | 11.7  | 10.5  | 11.5  | 14.9  | 17.2  | 19.9  | 20.7  | 20.3  | 18.2  | 16.9  | 14.2  | 12.1  |
|                                                                        | 5%          | DB     | 12.0  | 12.1  | 15.9  | 20.2  | 22.9  | 27.1  | 29.1  | 28.0  | 24.9  | 19.9  | 15.1  | 12.8  |
|                                                                        |             | MCWB   | 10.6  | 9.8   | 11.0  | 13.6  | 16.3  | 18.8  | 19.4  | 19.1  | 17.7  | 16.0  | 13.3  | 11.3  |
|                                                                        | 10%         | DB     | 10.9  | 11.1  | 13.8  | 17.9  | 20.8  | 24.9  | 27.0  | 25.6  | 22.9  | 18.1  | 14.0  | 11.8  |
|                                                                        |             | MCWB   | 9.5   | 9.0   | 10.2  | 12.2  | 15.3  | 17.9  | 18.8  | 17.9  | 16.7  | 15.2  | 12.3  | 10.4  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 12.9  | 12.1  | 13.6  | 17.3  | 20.3  | 23.5  | 22.7  | 22.4  | 20.7  | 19.0  | 15.7  | 13.6  |
|                                                                        |             | MCDB   | 13.5  | 14.4  | 17.9  | 22.7  | 26.1  | 32.5  | 32.1  | 31.4  | 25.6  | 22.1  | 17.0  | 14.1  |
|                                                                        | 2%          | WB     | 12.1  | 11.2  | 12.5  | 15.5  | 18.4  | 20.9  | 21.3  | 21.0  | 19.5  | 17.6  | 14.7  | 12.6  |
|                                                                        |             | MCDB   | 12.9  | 13.2  | 16.0  | 21.5  | 23.2  | 27.5  | 29.4  | 28.5  | 24.4  | 20.4  | 16.1  | 13.3  |
|                                                                        | 5%          | WB     | 11.0  | 10.4  | 11.6  | 14.2  | 16.9  | 19.6  | 20.3  | 19.9  | 18.5  | 16.7  | 13.6  | 11.5  |
|                                                                        |             | MCDB   | 12.1  | 11.9  | 14.9  | 19.8  | 21.5  | 26.0  | 27.4  | 25.6  | 23.2  | 19.2  | 15.2  | 12.6  |
|                                                                        | 10%         | WB     | 9.7   | 9.5   | 10.7  | 12.8  | 15.6  | 18.5  | 19.4  | 18.9  | 17.5  | 15.7  | 12.6  | 10.5  |
|                                                                        |             | MCDB   | 10.5  | 10.8  | 13.4  | 17.1  | 19.6  | 24.0  | 25.7  | 23.9  | 21.9  | 18.0  | 14.0  | 11.6  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 6.4   | 8.2   | 9.7   | 11.4  | 11.5  | 11.9  | 12.4  | 12.5  | 11.9  | 9.5   | 7.6   | 7.1   |
|                                                                        |             | MCDBR  | 6.3   | 9.4   | 13.2  | 14.9  | 15.1  | 15.1  | 15.6  | 16.9  | 14.9  | 10.8  | 8.0   | 7.3   |
|                                                                        |             | MCWBR  | 5.2   | 6.8   | 7.8   | 7.9   | 8.0   | 7.0   | 6.3   | 7.5   | 7.1   | 6.3   | 5.7   | 6.1   |
|                                                                        | 5% WB       | MCDBR  | 6.0   | 8.6   | 10.7  | 14.2  | 13.0  | 13.7  | 14.1  | 14.3  | 12.7  | 9.2   | 7.4   | 7.0   |
|                                                                        |             | MCWBR  | 5.4   | 6.7   | 6.9   | 8.0   | 7.9   | 7.0   | 6.4   | 7.5   | 7.2   | 6.0   | 5.8   | 6.3   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.326 | 0.344 | 0.373 | 0.395 | 0.399 | 0.408 | 0.402 | 0.390 | 0.367 | 0.362 | 0.335 | 0.321 |
|                                                                        | taud        |        | 2.439 | 2.384 | 2.300 | 2.242 | 2.279 | 2.287 | 2.305 | 2.356 | 2.399 | 2.419 | 2.439 | 2.415 |
|                                                                        | Ebn at Noon |        | 751   | 814   | 841   | 856   | 865   | 856   | 857   | 852   | 839   | 781   | 729   | 705   |
|                                                                        | Edh at Noon |        | 68    | 87    | 110   | 128   | 128   | 128   | 124   | 113   | 99    | 83    | 67    | 62    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.05  | 1.92  | 3.05  | 4.47  | 5.28  | 5.89  | 5.62  | 4.98  | 3.92  | 2.28  | 1.30  | 0.91  |
|                                                                        | RadStd      |        | 0.10  | 0.23  | 0.35  | 0.51  | 0.42  | 0.56  | 0.34  | 0.44  | 0.29  | 0.18  | 0.12  | 0.11  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (27 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page